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Importing Data with the RStudio IDE



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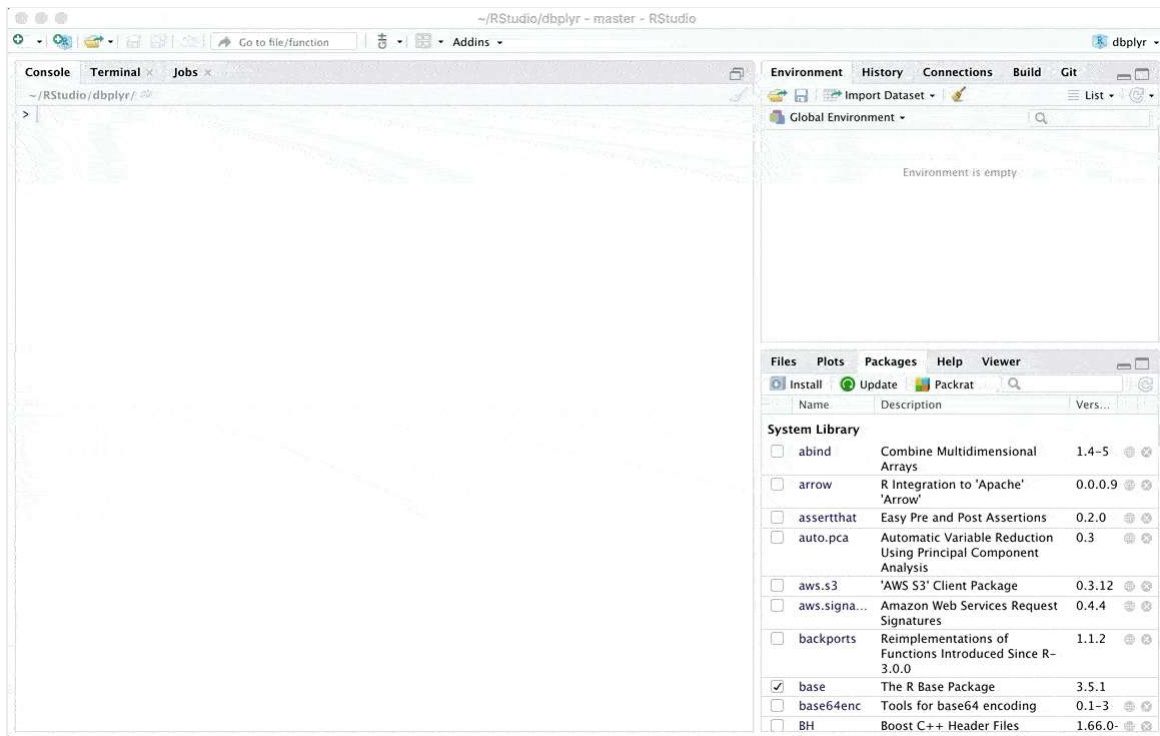
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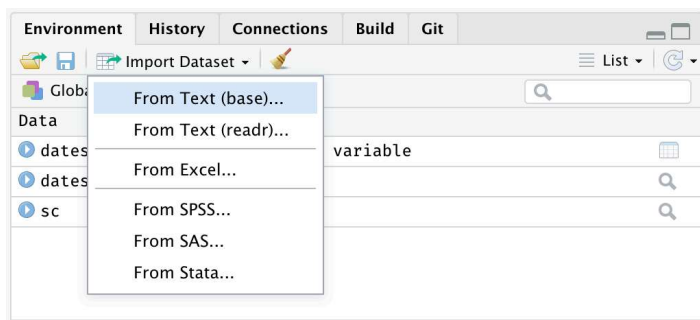
Introduction

Importing data into R is a necessary step that, at times, can become time intensive. To ease this task, the RStudio IDE includes new features to import data from: csv, xls, xlsx, sav, dta, por, sas and stata files.

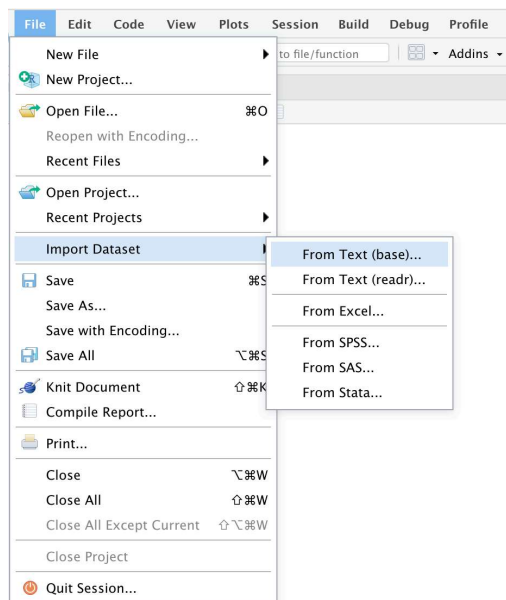


Importing data

The data import features can be accessed from the environment pane or from the tools menu. The importers are grouped into 3 categories: Text data, Excel data and statistical data. To access this feature, use the "Import Dataset" dropdown from the "Environment" pane:



Or through the "File" menu, followed by the "Import Dataset" submenu:



Importing data from Text and CSV files

Importing "From Text (readr)" files allows you to import CSV files and in general, character delimited files using the [readr](#) package. This Text importer provides support to:

- Import from the file system or a url
- Change column data types
- Skip or include-only columns
- Rename the data set
- Skip the first N rows
- Use the header row for column names
- Trim spaces in names
- Change the column delimiter
- Encoding selection
- Select quote, escape, comment and NA identifiers

For example, one can import with ease a csv form data.gov by pasting this url <https://data.montgomerycountymd.gov/api/views/2qd6-mr43/rows.csv?accessType=DOWNLOAD> and selecting "Import".

Import Text Data

File/Url: Update

Data Preview:

Full Name (character)	Gender (character)	Current Annual Salary (double)	2017 Gross Pay Received (double)	2017 Overtime Pay (double)	Department (character)	Department Name (character)	Division
Aarhus, Pam J.	F	70959.79	71316.72	0.00	POL	Department of Police	MSB Information Mgmt and Tech Division Rec
Aaron, Marsha M.	F	110359.00	108040.82	0.00	HHS	Department of Health and Human Services	Adult Protective and Case Management Servic
Abadio, Godfred A.	M	55950.24	62575.19	7649.19	COR	Correction and Rehabilitation	PRRS Facility and Security
Ababu, Essayas	M	95740.00	96055.94	0.00	HCA	Department of Housing and Community Affairs	Affordable Housing Programs
Abbamonte, Drew B.	M	74732.00	98736.78	23468.73	POL	Department of Police	PSB 6th District Special Assignment Team
Abbasian, Takin M.	M	16451.50	4547.10	94.92	POL	Department of Police	MSB Personnel Division
Abdalla, Eiman M.	F	63977.00	62177.20	184.56	FIN	Department of Finance	General Accounting
Abdelhamed, Shereen N.	F	53274.00	50549.01	1872.92	POL	Department of Police	PSB 2nd District Patrol
Abdelmoniem, Marwan M.	M	68739.36	65719.27	0.00	HHS	Department of Health and Human Services	Head Start
Abdool, Daniel Z.	M	50172.00	48567.94	1384.89	FRS	Fire and Rescue Services	Field Recruits
Abdul-Chani, Hacinah I.	F	50576.93	51657.55	293.30	POL	Department of Police	FSR Traffic Division Automated Traffic Enforce

Previewing first 50 entries.

Import Options:

Name: ☒ First Row as Names Delimiter: Escape:

Skip: ☒ Trim Spaces Quotes: Comment:

☒ Open Data Viewer Locale: NA:

Code Preview:

```
library(readr)
rows <- read_csv("https://data.montgomerycountymd.gov/api/views/2qd6-mr43/rows.csv?accessType=DOWNLOAD")
View(rows)
```

Reading rectangular data using readr Import Cancel

Importing data from Text files

Importing using "From Text (base)" enables importing text files using the base package, this is helpful to preserve compatibility with previous versions of RStudio.

Import Dataset

Name:

Input File:

Encoding:

Heading: ☒ Yes ☐ No

Row names:

Separator:

Decimal:

Quote:

Comment:

na.strings:

☒ Strings as factors

Data Frame:

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.4	3.9	1.7	0.4	setosa
4.6	3.4	1.4	0.3	setosa
5.0	3.4	1.5	0.2	setosa
4.4	2.9	1.4	0.2	setosa
4.9	3.1	1.5	0.1	setosa
5.4	3.7	1.5	0.2	setosa
4.8	3.4	1.6	0.2	setosa
4.8	3.0	1.4	0.1	setosa

Import Cancel

Importing data from Excel files

The Excel importer provides support to:

- Import from the file system or a url

- Change column data types
- Skip columns
- Rename the data set
- Select an specific Excel sheet
- Skip the first N rows
- Select NA identifiers

For example, one can import with ease an xls file from data.gov by pasting this url <http://www.fns.usda.gov/sites/default/files/pd/slsummar.xls> and selecting "Update".

Notice that this file contains to tables and therefore, requires the first few rows to be removed.

Import Excel Data

File/Url:
 Update

Data Preview:

NATIONAL SCHOOL LUNCH PROGRAM: PARTICIPATION AND LUNCHES SERVED	X_1 (character)	X_2 (character)	X_3 (character)	X_4 (character)	X_5 (character)	X_6 (character)
(Data as of October 05, 2018)	NA	NA	NA	NA	NA	NA
-----Average Participation-----	NA	NA	NA	NA	NA	NA
Fiscal Year	Free	Reduced	Full	Total	Total Lunches Served	Percent Free/RP of Total
	-----Millions-----	NA	NA	NA	NA	%
1969	2.899999999999999	1]	16.5	19.399999999999999	3368.1999999999998	15.1
1970	4.5999999999999996	1]	17.800000000000001	22.399999999999999	3565.0999999999999	20.699999999999999
1971	5.7999999999999998	0.5	17.800000000000001	24.100000000000001	3848.3000000000002	26.100000000000001
1972	7.2999999999999998	0.5	16.600000000000001	24.399999999999999	3972.0999999999999	32.399999999999999

Previewing first 50 entries.

Import Options:

Name: Max Rows: ☒ First Row as Names

Sheet: Skip: ☒ Open Data Viewer

Range: NA:

Code Preview:

```
library(readxl)
url <- "http://www.fns.usda.gov/sites/default/files/pd/slsummar.xls"
destfile <- "slsummar.xls"
curl::curl_download(url, destfile)
slsummar <- read_excel(destfile)
View(slsummar)
```

Reading Excel files using readxl Import Cancel

We can clean this up by skipping 6 rows from this file and unchecking the "First Row as Names" checkbox.

Import Excel Data

File/Url: Update

Data Preview:

X_1 (character)	X_2 (double)	X_3 (character)	X_4 (double)	X_5 (double)	X_6 (double)	X_7 (character)
1969	2.9	1]	16.5	19.4	3368.20	15.1
1970	4.6	1]	17.8	22.4	3565.10	20.699999999999999
1971	5.8	0.5	17.8	24.1	3848.30	26.100000000000001
1972	7.3	0.5	16.6	24.4	3972.10	32.399999999999999
1973	8.1	0.5	16.1	24.7	4008.80	35
1974	8.6	0.5	15.5	24.6	3981.60	37.100000000000001
1975	9.4	0.5999999999999998	14.9	24.9	4063.00	40.299999999999997
1976	10.2	0.8000000000000004	14.6	25.6	4147.90	43.100000000000001
1977	10.5	1.3	14.5	26.2	4250.00	44.799999999999997
1978	10.3	1.5	14.9	26.7	4294.10	44.399999999999999
1979	10.0	1.7	15.3	27.0	4357.40	43.600000000000001
1980	10.0	1.8999999999999999	14.7	26.6	4387.00	45.100000000000001

Previewing first 50 entries.

Import Options:

Name: Max Rows: ☐ First Row as Names

Sheet: Skip: ☒ Open Data Viewer

Range: NA:

Code Preview:

```
library(readxl)
url <- "http://www.fns.usda.gov/sites/default/files/pd/slsummar.xls"
destfile <- "slsummar.xls"
curl::curl_download(url, destfile)
slsummar <- read_excel(destfile, col_names = FALSE,
  skip = 6)
```

[Reading Excel files using readxl](#)

Import Cancel

The file is looking better but some columns are being displayed as strings when they are clearly numerical data. We can fix this by selecting "numeric" from the column dropdown.

Import Excel Data

File/Url: Update

Data Preview:

X_1 (numeric)	X_2 (double)	X_3 (numeric)	X_4 (double)	X_5 (double)	X_6 (double)	X_7 (numeric)
1969	2.9	NA	16.5	19.4	3368.20	15.1
1970	4.6	NA	17.8	22.4	3565.10	20.7
1971	5.8	0.5	17.8	24.1	3848.30	26.1
1972	7.3	0.5	16.6	24.4	3972.10	32.4
1973	8.1	0.5	16.1	24.7	4008.80	35.0
1974	8.6	0.5	15.5	24.6	3981.60	37.1
1975	9.4	0.6	14.9	24.9	4063.00	40.3
1976	10.2	0.8	14.6	25.6	4147.90	43.1
1977	10.5	1.3	14.5	26.2	4250.00	44.8
1978	10.3	1.5	14.9	26.7	4294.10	44.4
1979	10.0	1.7	15.3	27.0	4357.40	43.6
1980	10.0	1.9	14.7	26.6	4387.00	45.1

Previewing first 50 entries.

Import Options:

Name: Max Rows: ☐ First Row as Names

Sheet: Skip: ☒ Open Data Viewer

Range: NA:

Code Preview:

```
library(readxl)
url <- "http://www.fns.usda.gov/sites/default/files/pd/slsummar.xls"
destfile <- "slsummar.xls"
curl::curl_download(url, destfile)
slsummar <- read_excel(destfile, col_names = FALSE,
  col_types = c("numeric", "numeric", "numeric",
```

[Reading Excel files using readxl](#)

Import Cancel

The final step is to click "Import" to run the code under "Code Preview" and import the data into RStudio, the final result should look as follows:

The screenshot shows the RStudio IDE interface. The main window displays a data frame named 'slsummar' with 53 entries. The columns are labeled X_1 through X_7. The console shows the following R code:

```
> library(readxl)
> url <- "http://www.fns.usda.gov/sites/default/files/pd/slsummar.xls"
> destfile <- "slsummar.xls"
> curl::curl_download(url, destfile)
> slsummar <- read_excel(destfile, col_names = FALSE,
+   col_types = c("numeric", "numeric", "numeric",
+     "numeric", "numeric", "numeric", "numeric",
+     "numeric"), skip = 6)
There were 50 or more warnings (use warnings() to see the first 50)
> View(slsummar)
```

The Environment pane on the right shows the 'slsummar' dataset with 53 observations and 7 variables. The 'destfile' is 'slsummar.xls' and the 'url' is 'http://www.fns.usda.gov/sites/d...'. The Packages pane shows the installed packages, including 'base' and 'base64enc'.

Importing data from SPSS, SAS and Stata files

The SPSS, SAS and Stata importer provides support to:

- Import from the file system or a url
- Rename the data set
- Specify a model file

We can import https://github.com/rstudio/webinars/raw/master/23-Importing-Data-into-R/data/Child_Data.sav by pasting the address under File/Url and clicking "Update" followed by clicking "Import".

The 'Import Statistical Data' dialog box is shown. The 'File/Url' field contains the URL: https://github.com/rstudio/webinars/raw/master/23-Importing-Data-into-R/data/Child_Data.sav. The 'Data Preview' section shows a table with the following columns: AGE, MEM_SPAN, IQ, and READ_AB. The 'Import Options' section shows the Name: 'Child_Data', Model: (empty), and Format: 'SAV'. The 'Code Preview' section shows the following R code:

```
library(haven)
Child_Data <- read_sav("https://github.com/rstudio/webinars/raw/master/23-Importing-Data-into-R/data/Child_Data.sav")
View(Child_Data)
```

The 'Import' button is highlighted in blue.

Need Help?

RStudio Pro customers may open a discussion with RStudio Support at any time.

You may also ask for help from R and RStudio users on community.rstudio.com. Be sure to include a reproducible example of your issue. [Click here to start a new community discussion.](#)



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